

# Research Proposal: Unifying Continuous-Time State-Space Models with Formal Methods for Transparent NLU

## Context and Motivation

Large Language Models (LLMs) excel at predictive text generation but fundamentally lack structural grounding, leading to critical issues with “hallucinations” and a failure to generalize on structured reasoning tasks. While the NLP field increasingly relies on scale (more parameters, more data) to overcome these limitations, I propose a fundamental architectural shift. Instead of treating language merely as a discrete probabilistic distribution of tokens, we can model it as a formal dynamical system governed by strict logic. By integrating continuous-time state-space models (which I have previously applied to physical systems) with symbolic reasoning constraints, we can achieve transparent, efficient, and bounded Natural Language Understanding (NLU).

## Proposed Methodology

My research will focus on combining neural architectures with logical methods to define task-relevant constraints on model generation. I propose a three-tier framework:

1. **State-Space Language Modeling (SSLM):** Building upon my previous work with Medical\_MambaTS, I will investigate the application of linear-time state-space models (SSMs) to natural language. Unlike standard transformer attention mechanisms, SSMs rely on continuous-time differential equations to map sequences. This mathematical structure is naturally conducive to formal verification because the transition matrices can be analytically bounded, allowing us to mathematically prove the model’s stability during long-context reasoning tasks.
2. **Logical Constraint Enforcement via Autograd (Neuro-Symbolic Integration):** Drawing from my experience engineering Physics-Informed Operator Networks (PI-DeepONet), I will develop a mechanism to enforce formal logical constraints directly within the neural network’s loss landscape. Instead of enforcing physical laws (e.g., fluid dynamics), the network will be constrained by formal grammars, type theory, and First-Order Logic (FOL). If the model generates a sequence that violates logical consistency (a hallucination or a reasoning flaw), a “Logic-Informed Loss” function will heavily penalize the gradient during backpropagation. This ensures the model learns to respect structural rules, drastically improving generalization on reasoning tasks without requiring exponential data scaling.
3. **Probing and Verification in Ambiguous Contexts:** I will develop probing mechanisms to extract the latent state representations of the SSLM. By applying algebraic topology and formal verification techniques

to these latent manifolds, we can verify whether the model is genuinely executing deductive logic or merely relying on associative heuristics. This will provide unprecedented transparency into the model’s decision-making process, especially when evaluating NLU models in the presence of linguistic ambiguity or conflicting annotator data.

## **Expected Impact**

By embedding formal methods directly into the architecture and training loop of state-space language models, this project will yield LLMs that are not only vastly more computationally efficient ( $O(N)$  complexity) but inherently trustworthy. This neuro-symbolic approach will allow us to mathematically guarantee reasoning bounds, paving the way for the safe deployment of language models in high-stakes environments.